Practice workbook · Willingham's learning theory

Factual knowledge before skill, practice and effort, and strategies for cognitive, physical and social development.

9 question parts: 1 worked-example retry and 8 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: PSY-DEV-04

1. Question 4 · part b · 2 marks

2020 NOV · Paper 01

Worked-example retry · Attempt part b only.

4 Mr Harris's class has begun to learn about the Solar System.

He collects data on how the children in his class perform in attempting to name the order of the planets correctly in terms of their distance from the Sun.

Mr Harris tests them at the end of each day on four separate days.

A summary of Mr Harris's class's performance is shown in **Table 1**.

	Day 1 test	Day 2 test	Day 3 test	Day 4 test
Average percentage of planets placed in the correct order	32%	32%	33%	33%
Mean time taken to complete the test (in seconds)	21.9	21.0	20.8	21.2

Table 1

(a) Explain **two** conclusions that can be made using the data in **Table 1** regarding Mr Harris's class's performance on the test.

(4)

Conclusion 1

Conclusion 2

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1. Question 4 · part b · 2 marks

2020 NOV · Paper 01

Worked-example retry · Attempt part b only.

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(b) Explain how Mr Harris could help his class improve their performance on the test.

You should refer to Daniel Willingham's learning theory in your answer.

(2)

(Total for Question 4 = 6 marks)

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Working and answer

2. Question 3 · part a · 2 marks

2019 JUN · Paper 01

Further practice · Attempt part a only.

3 Mario is a teacher who works at a primary school. His new class of pupils are struggling with their physical development.

(a) Explain, using Daniel Willingham's learning theory, **one** way Mario could support his pupils' physical development.

(2)

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2. Question 3 · part a · 2 marks

2019 JUN · Paper 01

Further practice · Attempt part a only.

(b) Explain **one** strength and **one** weakness of using Daniel Willingham's learning theory with Mario's pupils.

(4)

Strength:

Weakness:

(Total for Question 3 = 6 marks)

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2. Question 3 · part a · 2 marks · Working and answer

Working and answer

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3. Question 3 · part b · 4 marks

2019 JUN · Paper 01

Further practice · Attempt part b only.

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3 Mario is a teacher who works at a primary school. His new class of pupils are struggling with their physical development.

(a) Explain, using Daniel Willingham's learning theory, **one** way Mario could support his pupils' physical development.

(2)

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3. Question 3 · part b · 4 marks

2019 JUN · Paper 01

Further practice · Attempt part b only.

(b) Explain **one** strength and **one** weakness of using Daniel Willingham's learning theory with Mario's pupils.

(4)

Strength:

Weakness:

(Total for Question 3 = 6 marks)

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3. Question 3 · part b · 4 marks · Working and answer

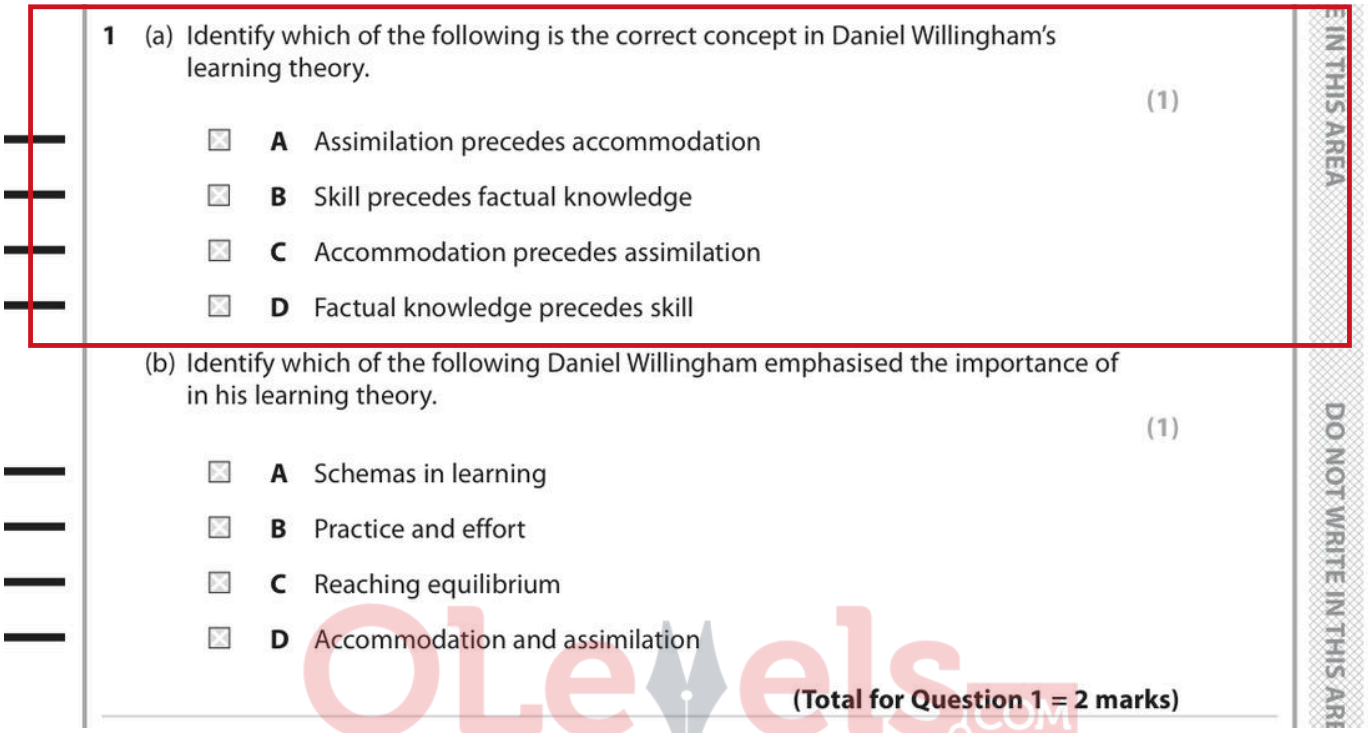
Working and answer

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4. Question 1 · part a · 1 mark

2021 NOV · Paper 01

Further practice · Attempt part a only.



1 (a) Identify which of the following is the correct concept in Daniel Willingham's learning theory.

(1)

- ☒ **A** Assimilation precedes accommodation
- ☒ **B** Skill precedes factual knowledge
- ☒ **C** Accommodation precedes assimilation
- ☒ **D** Factual knowledge precedes skill

(b) Identify which of the following Daniel Willingham emphasised the importance of in his learning theory.

(1)

- ☒ **A** Schemas in learning
- ☒ **B** Practice and effort
- ☒ **C** Reaching equilibrium
- ☒ **D** Accommodation and assimilation

(Total for Question 1 = 2 marks)

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Working and answer

5. Question 1 · part b · 1 mark

2021 NOV · Paper 01

Further practice · Attempt part b only.

1 (a) Identify which of the following is the correct concept in Daniel Willingham's learning theory.

(1)

- ☒ **A** Assimilation precedes accommodation
- ☒ **B** Skill precedes factual knowledge
- ☒ **C** Accommodation precedes assimilation
- ☒ **D** Factual knowledge precedes skill

(b) Identify which of the following Daniel Willingham emphasised the importance of in his learning theory.

(1)

- ☒ **A** Schemas in learning
- ☒ **B** Practice and effort
- ☒ **C** Reaching equilibrium
- ☒ **D** Accommodation and assimilation

(Total for Question 1 = 2 marks)

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Working and answer

6. Question 3 · part whole · 2 marks

2021 NOV · Paper 01

Further practice · Attempt part whole only.

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3 Molly and Mack are struggling with their social development at school.

Their teacher, Mrs Kumar, has noticed that Molly and Mack snatch toys from other children when they want to play with them. They push other children in the playground and shout in their faces when they are eating their lunch.

Mrs Kumar wants to help improve Molly and Mack's social development.

Explain **one** strategy that Mrs Kumar could use to help support the social development of Molly and Mack.

You should refer to Daniel Willingham's learning theory in your answer.

(Total for Question 3 = 2 marks)

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Working and answer

7. Question 5 · part a · 2 marks

2022 JUN · Paper 01

Further practice · Attempt part a only.

5 James is a teacher at a school. He wants to see whether social learning can be used to help the social development of his pupils.

He splits a group of 10 pupils in half, keeping five of the pupils in his classroom while another teacher takes the other five pupils to their classroom. James shows them how to share and take turns on a scooter and asks them to practise this whilst he watches and supervises them. The other teacher just gives the pupils a scooter and supervises only. This is done once a day for a week.

The following week James observes his whole class and gives each of the 10 pupils a score for their ability to share and take turns. The pupils were scored from 1 to 10, where 1 indicated very poor ability to share and take turns and 10 indicated excellent ability to share and take turns.

Table 1 shows the scores given to each pupil by James.

Supervised by James		Supervised by the other teacher	
Pupil	**Ability to share and take turns (1–10)**	**Pupil**	**Ability to share and take turns (1–10)**
A	8	F	4
B	9	G	5
C	8	H	5
D	7	I	4
E	9	J	6

Table 1

(a) Explain **one** conclusion you could make from **Table 1** regarding social learning to improve social development.

(2)

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P 6 8 8 2 3 R A 0 6 4 0

7. Question 5 · part a · 2 marks

2022 JUN · Paper 01

Further practice · Attempt part a only.

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(b) Explain **one** improvement that James could have made to his investigation.

(2)

(Total for Question 5 = 4 marks)

Working and answer

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8. Question 4 · part a · 4 marks

2023 JUN · Paper 01

Further practice · Attempt part a only.

4 Sergio wanted to see the influence of practice and effort on learning a new skill.

Sergio decided to show his niece and nephew how to juggle one weekend. He asked his niece to practise juggling every day for a week and when she sent him videos of her practising juggling, he sent her supportive messages and praised her effort. Sergio did not ask his nephew to practise juggling and did not send him any supportive messages.

Sergio was sent videos of his niece and nephew juggling at the beginning and the end of the week. He timed how long they could juggle for without dropping the juggling balls.

His results are shown in **Figure 2**.

A comparative bar chart to show the time (in seconds) that the niece and nephew were able to juggle at the beginning and end of the week

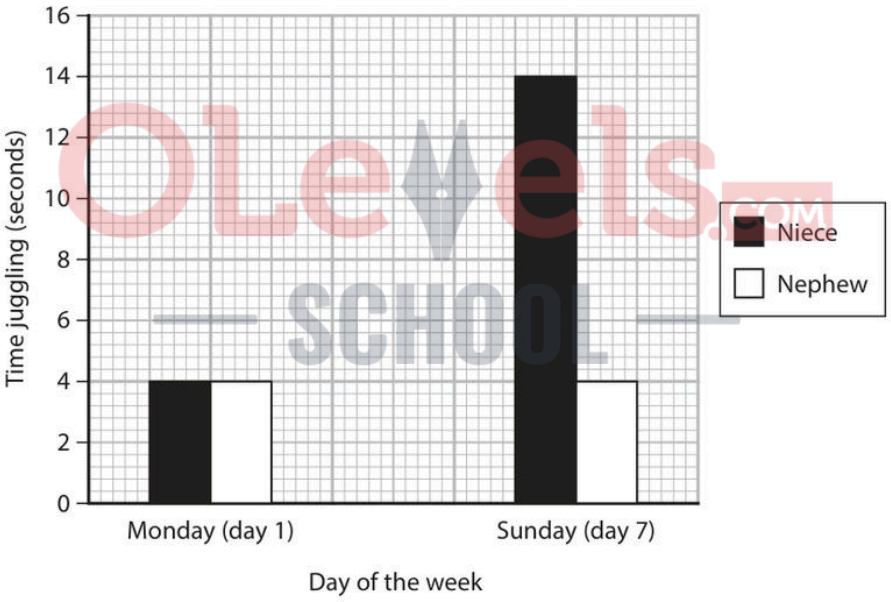


Figure 2

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8. Question 4 · part a · 4 marks

2023 JUN · Paper 01

Further practice · Attempt part a only.

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(a) Explain **two** conclusions that could be made from **Figure 2** regarding the role of practice and effort in learning a new skill.

(4)

1 .....

2 .....

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P 7 1 2 8 6 A 0 7 3 6

7

Turn over ▶

8. Question 4 · part a · 4 marks

2023 JUN · Paper 01

Further practice · Attempt part a only.

(b) Explain **two** weaknesses of Sergio's investigation into the role of practice and effort in learning a new skill.

(4)

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2 .....

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(Total for Question 4 = 8 marks)

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8. Question 4 · part a · 4 marks · Working and answer

Working and answer

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9. Question 3 · part a · 4 marks

2025 JUN · Paper 01

Further practice · Attempt part a only.

3 Alessia is a teacher at a school and is trying to help students learn a new concept in science class.

Alessia split her students into three groups to see who would learn the new concept the most effectively.

- Group 1 – the students were asked to do a very easy task about the new concept, with only very simple activities included.
- Group 2 – the students were asked to do a reasonably difficult task about the new concept, with some slightly more challenging activities included.
- Group 3 – the students were asked to do a very hard, complex task about the new concept, with only extremely challenging activities included.

Alessia asked her students to do the activities in their groups over the course of a week during lessons. At the start and the end of the week the students took the same test about the new science concept.

Alessia wanted to make some conclusions about how the tasks given affected the students' performance in the test.

Figure 1 shows the average performance of the three groups at the start and the end of the week.

A comparative bar chart to show the average test performance (%) on the new science concept at the start and the end of the week

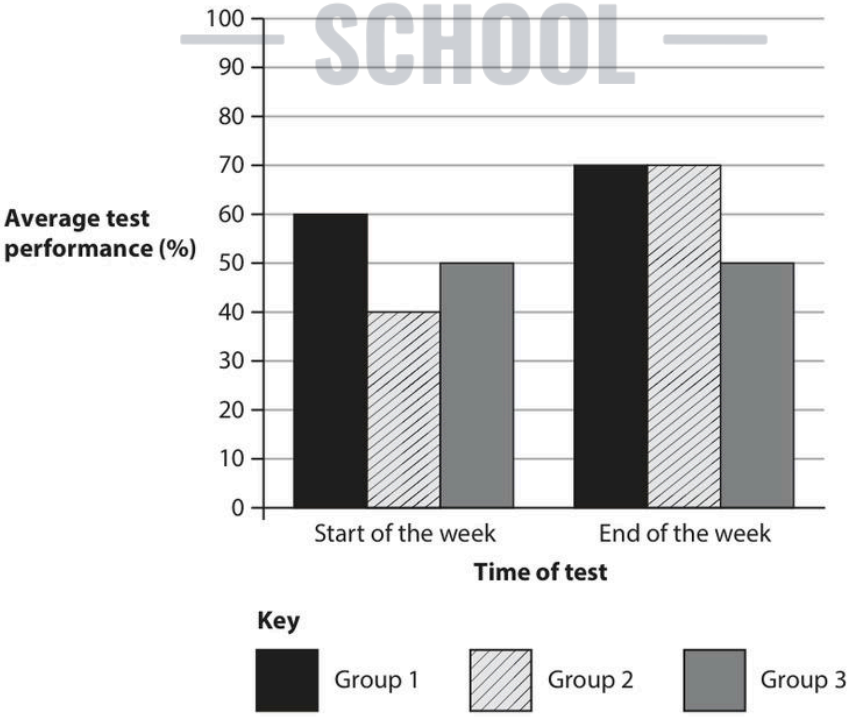


Figure 1

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P 7 8 7 7 2 A 0 4 4 0

9. Question 3 · part a · 4 marks

2025 JUN · Paper 01

Further practice · Attempt part a only.

(a) Explain **two** conclusions that could be made using the data in **Figure 1** in terms of how appropriate the tasks given to the three groups of students were in learning a new concept.

(4)

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2 .....

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P 7 8 7 7 2 A 0 5 4 0

9. Question 3 · part a · 4 marks

2025 JUN · Paper 01

Further practice · Attempt part a only.

(b) Explain **one** strength and **one** weakness of Alessia's investigation into learning a new concept.

(4)

Strength

Weakness

(Total for Question 3 = 8 marks)

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9. Question 3 · part a · 4 marks · Working and answer

Working and answer

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